

Dedai Wei

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Google Scholar: <https://scholar.google.com/citations?hl=zh-CN&user=A74U1IYAAAAJ>

EDUCATION

University of Macau — Macau, China

08/2025 – 07/2027 (Expected)

Master of Economics, Major: Economics

TOEFL: 103/120

Core Courses: Microeconomic Theory, Macroeconomic Theory, Methods of Economic Research, Econometric Analysis, International Trade and Finance, Development Economics, Public Economics, Labour Economics.

Shenyang University — Shenyang, China

09/2018 – 06/2022

Bachelor of Economics, Major: Statistics in Economics

Core Courses: Fundamentals of Statistics, Multivariate Statistical Analysis, SPSS Software Applications, Econometrics, Time Series Analysis, Sampling Techniques Survey.

Research Interests: Neural Networks, Probabilistic Diffusion Modeling, Transfer Learning, Financial Machine Learning, and High-Dimensional Econometrics.

Skills: Python, R, MATLAB, Stata, MS Office, LaTeX

PUBLICATIONS

Published / Accepted

- [1] **Wei, D.**, Ouyang, K., et al. (2026). "Multi-strategy enhanced dung beetle algorithm and its application for bankruptcy prediction." *Neural Networks* (JCR Q1, IF 6.3).
- [2] Yuan, Y., **Wei, D.**, et al. (2026). "Hedging the Black Box: A Feasibility Study of Financial Risk Transfer for AI Adoption via Generative Agent Simulation." *Workshop at the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)* (Online).
- [3] **Wei, D.**, et al. (2025). "A comprehensive analysis of digital inclusive finance's influence on high quality enterprise development through fixed effects and deep learning frameworks." *Scientific Reports* (JCR Q1, IF 4.3).
- [4] Wan, M., **Wei, D.**, et al. (2025). "Low-carbon Pilot Cities and Financialization of Listed Companies: Inhibitory Effects and Policy Analysis." *Finance Research Letters* (JCR Q1, IF 7.6).
- [5] Ouyang, K., **Wei, D.**, et al. (2025). "Beaver Behavior optimizer: a novel metaheuristic algorithm for solar PV parameter identification and engineering problems." *Journal of Advanced Research* (JCR Q1, IF 13).
- [6] **Wei, D.**, et al. (2025). "Multiple Objectives Escaping Bird Search Optimization and Its application in Stock Market Prediction Based on Transformer Model." *Scientific Reports* (JCR Q1, IF 4.3).
- [7] Ouyang, K., **Wei, D.**, et al. (2025). "Multi-objective Red-billed Blue Magpie Optimizer: A Novel Algorithm for Multi-objective UAV Path Planning." *Results in Engineering* (JCR Q1, IF 7.9).
- [8] Xiao, W., **Wei, D.**, et al. (2025). "Newton Downhill Optimizer with Application to Engineering Optimization and Breast Cancer Feature Selection." *Biomedical Signal Processing and Control* (JCR Q1, IF 4.9).
- [9] Xiao, W., Ouyang, K., . . . , **Wei, D.**, et al. (2025). "FLNM-Net: A Frequency-Adaptive and Luminance-Noise Aware Mask Network for rPPG Signal Extraction from Video." *Pattern Recognition* (JCR Q1, IF 7.5).

Under Review

- [1] Yuan, Y., **Wei, D.**, et al. (2026). "Hedging the Black Box: A Feasibility Study of Financial Risk Transfer for AI Adoption via Generative Agent Simulation." *IEEE Transactions on Artificial Intelligence* (Scopus Q1, CiteScore 6.4; Under Review).
- [2] **Wei, D.**, et al. (2026). "PIC-GO: Core-First Residual Routing for Runtime-Hidden Physical Regime." *AAAI 2027* (Under Review).
- [3] **Wei, D.**, et al. (2026). "StressGDiff: Risk-Guided Heterogeneous Graph Diffusion for Stress-Scenario Discovery in Renewable-Rich Power Systems." *Reliability Engineering & System Safety* (JCR Q1, IF 13.7; Under Review).
- [4] **Wei, D.**, et al. (2026). "Text-Enhanced Dynamic Multi-View Graph Diffusion Networks with LLM-Based Validation for Financial Structural Breakpoint Detection." *Information Sciences* (JCR Q1, IF 6.8; Under Review).

Working Papers

- [1] **Wei, D.** (2026). "Rank-Gated Neural Estimation of Latent Equilibrium Spaces in Nonstationary Multivariate Time Series." Working Paper.
- [2] **Wei, D.** (2026). "EI-HLRD: Engineering-Informed Hierarchical Latent Residual Diffusion for Probabilistic NGA-Sub Ground-Motion Modeling and Cross-Regional Transfer." Working Paper.
- [3] **Wei, D.** (2026). "FADC: LLM-Compiled Control Specifications for Intent-Aligned Subgroup Discovery." Working Paper.
- [4] **Wei, D.** (2026). "CausalMB-LLM: LLM-Guided Causal Relation and Markov Boundary Discovery for Tabular Prediction." Working Paper.

REVIEWER EXPERIENCE

[1] *Scientific Reports* (JCR Q1, IF 3.9; 5-year IF 4.3) – 2 reviews

[2] *Discover Artificial Intelligence* (CiteScore 6.0) – 2 reviews

[3] *Journal of King Saud University - Computer and Information Sciences* (JCR Q1, IF 6.1) – 1 review

RESEARCH EXPERIENCE

High-Dimensional Statistics, Structural Breaks, and Stochastic Processes

Research Assistant, University of Macau / Oct. 2025–Dec. 2025

- Structural-Break and Stochastic-Volatility Modelling:** Built a $p \gg n$ pipeline using SICS, CCA, RRS/BIC break detection, and regime-specific selection; modeled high-frequency returns with Heston jump-diffusion and OHLC detrending. (Working Paper)

Physics-Informed Neural Networks and Non-Stationary Dynamic Modelling

- CoEC-Net: Rank-Gated Equilibrium Learning:** Designed a neural error-correction model with multiscale memory, rank gates, low-rank bases, and equilibrium adjustment; recovered designed ranks and strong forecasts across nine datasets. (Working Paper)
- EI-HLRD: Latent Residual Diffusion for Seismic Spectra:** Built a probabilistic NGA-Sub workflow with teacher-student center prediction, event/within-event residuals, latent diffusion, DDIM sampling, calibration, and CORAL regional transfer. (Working Paper)
- PIC-GO: Gated Physics-Informed Neural Operator:** Combined a frozen flux-form core with residual gates for selective correction under local mechanism change; reduced mean relative L_2 error by 9.67% on AdvDiff-2D. (Under Review at AAAI 2027)

Dynamic Multi-Graph Learning and Structural Change Identification

- Text-Enhanced Dynamic Graph Diffusion:** Fused numerical, contextual, and text-risk signals to detect financial breakpoints via prediction drift, topology change, latent shifts, and uncertainty; interpreted regimes with GAM-DML. (Under Review at *Information Sciences*)
- Dynamic Multi-Graph Topological Shift Modelling:** Built dynamic graphs for time-varying nonlinear dependencies and tracked changes in connectivity, centrality, and propagation structure.

Controllable Large Language Models and Structured Discovery

- Controllable LLMs for Structured Discovery:** Built FADC for validated typed controls in subgroup discovery and extended it with LLM-guided Markov Boundary screening for interpretable tabular prediction. (Working Paper)

HONORS

National Challenge of Mathematical Ability in HEIs (Excellence Award)	2023
National Economic Virtual Imitation Experiment Competition (Provincial 2nd Prize)	2022
University Students Market Research and Analysis Competition (Provincial 3rd Prize)	2020
CFCC Smart Economy Innovation and Entrepreneurship Competition (Provincial 1st Prize)	2020
Certificate of ACCA Outstanding Campus Ambassador	2021
Certificate of UN ESG Green Economy Program (Sustainable Finance)	2023
HSBC Finance Day Learning Certificate	2023